CNENGR202 Instrumentation and Control

Part 1 L9 – Feedback Control - frequency domain

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Class question 1

$$G(s) = \frac{1}{1+3s}$$

- Find M and $\arg(G(\omega))$

Solution

$$G(j\omega) = \frac{1}{1+ j3\omega}$$

$$M = \frac{1}{|1+ j3\omega|} = \frac{1}{\sqrt{1+9\omega^2}}$$

$$\phi = \arg\left(\frac{1}{1+ j3\omega}\right) = \arg(1) - \arg(1+ j3\omega)$$

$$= \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{3\omega}{1}\right)$$

$$= \tan^{-1}(3\omega)$$

Numerical

Frequency ω	Magnitude M	Magnitude $\log_{10} M$	Phase(rads) ϕ	Phase (degrees)
0	1	0	0	0
1/3	0.7071	-0.15	-0.79	-45
1	0.3162	-0.50	-1.25	-72
3	0.1104	-0.96	-1.46	-84
10	0.0333	-1.48	-1.54	-88
∞	0	$-\infty$	-1.57	-90

Class question 2

$$G(s) = \frac{1}{(s+1)(s+5)}$$

- Find M and $\arg(G(\omega))$

Solution

$$G(j\omega)=\frac{1}{(1+j\omega)(5+j\omega)}$$

$$M=\frac{1}{|1+j\omega||5+j\omega|}=\frac{1}{\sqrt{(1+\omega^2)(25+\omega^2)}}$$

$$\phi=\arg\left(\frac{1}{(1+j\omega)(5+j\omega)}\right)=\arg(1)-\arg(1+j\omega)-\arg(5+j\omega)$$

$$=-\tan^{-1}(\omega)-\tan^{-1}\left(\frac{\omega}{5}\right)$$

Numerical

Frequency ω	Magnitude M	Magnitude $\log_{10} M$	Phase(rads) ϕ	Phase (degrees)
0	0.2	-0.70	0	0
1	0.1387	-0.86	-0.98	-56
5	0.0277	-1.56	-2.16	-124
10	0.0089	-2.05	-2.58	-148
100	0.0001	-4.000	-3.08	-177
∞	0	$-\infty$	-1.57	-180

Class question 3

$$G(s)=\frac{1}{(s+4)(s+5)(s+7)}$$

- Find M and $\arg(G(\omega))$

Solution

$$G(s)=\frac{1}{(s+4)(s+5)(s+7)} \qquad G(\omega)=\frac{1}{(j\omega+4)(j\omega+5)(j\omega+7)}$$

$$\left|G(\omega)\right|=M=\frac{1}{\sqrt{(\omega^2+16)(\omega^2+25)(\omega^2+49)}}$$

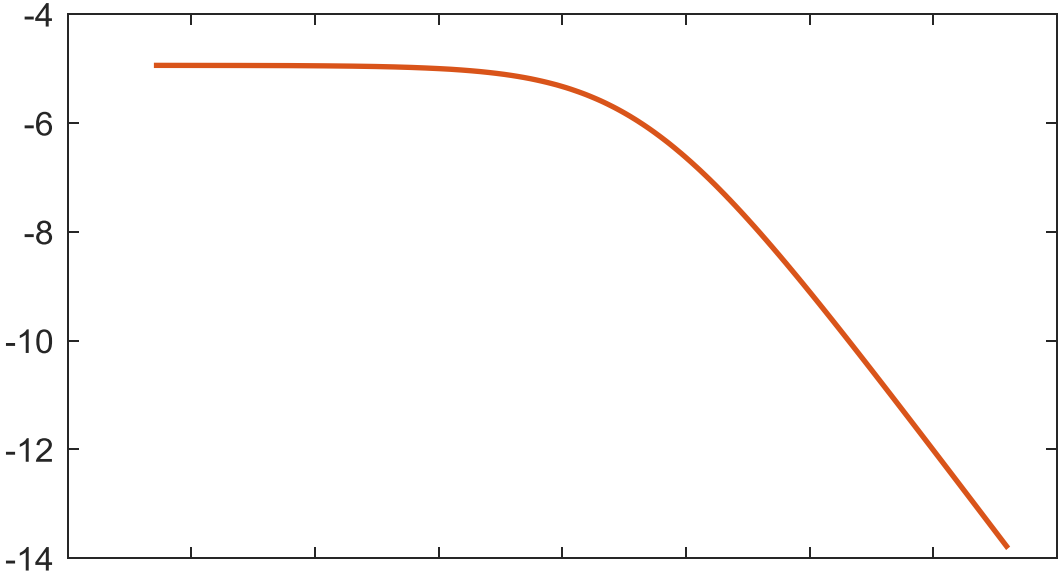
$$Arg(G(\omega))=-Arg(4+j\omega)-Arg(5+j\omega)-Arg(7+j\omega)$$

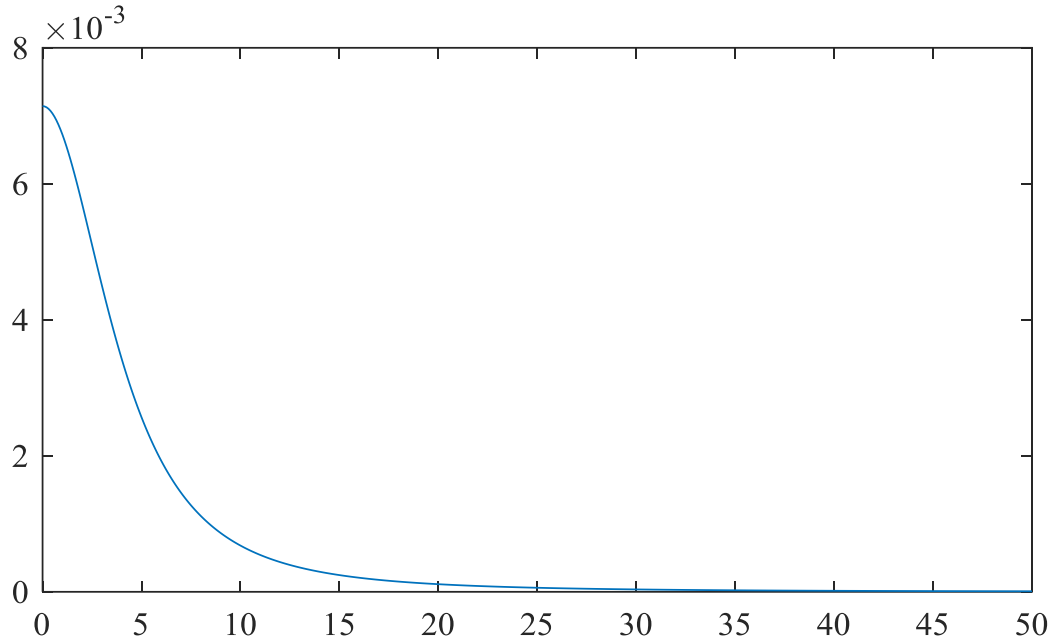
$$=-\tan^{-1}\left(\frac{\omega}{4}\right)-\tan^{-1}\left(\frac{\omega}{5}\right)-\tan^{-1}\left(\frac{\omega}{7}\right)$$

Numerical

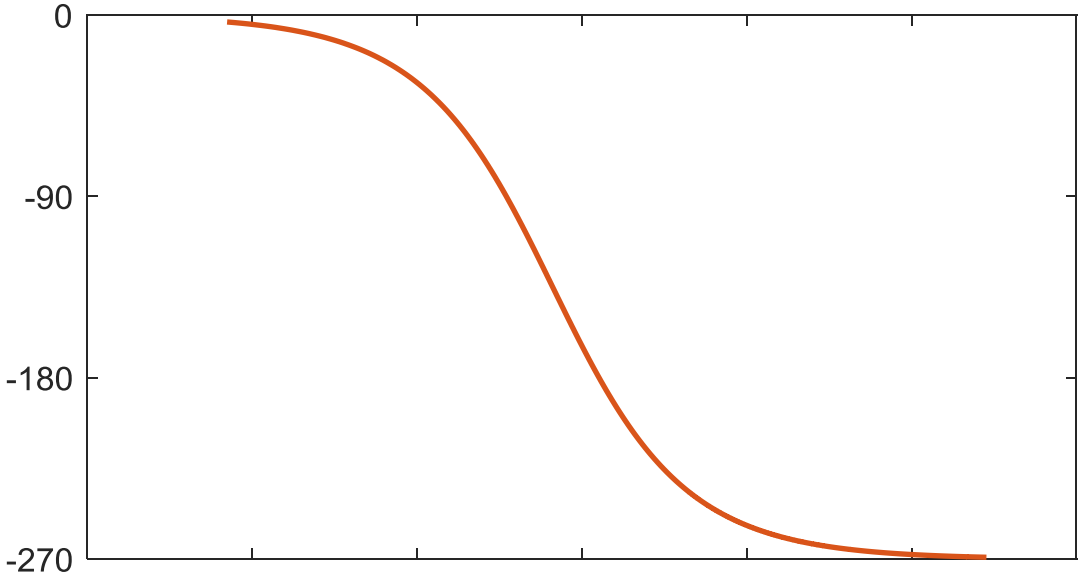
Frequency ω	**Magnitude** M	**Phase (rads)** ϕ	**Phase (degrees)**
0	0.0071	0	0
1	0.0071	-0.20	-11
3	0.0067	-0.58	-34
10	0.0007	-3.26	-187
∞	0	-4.71	-270

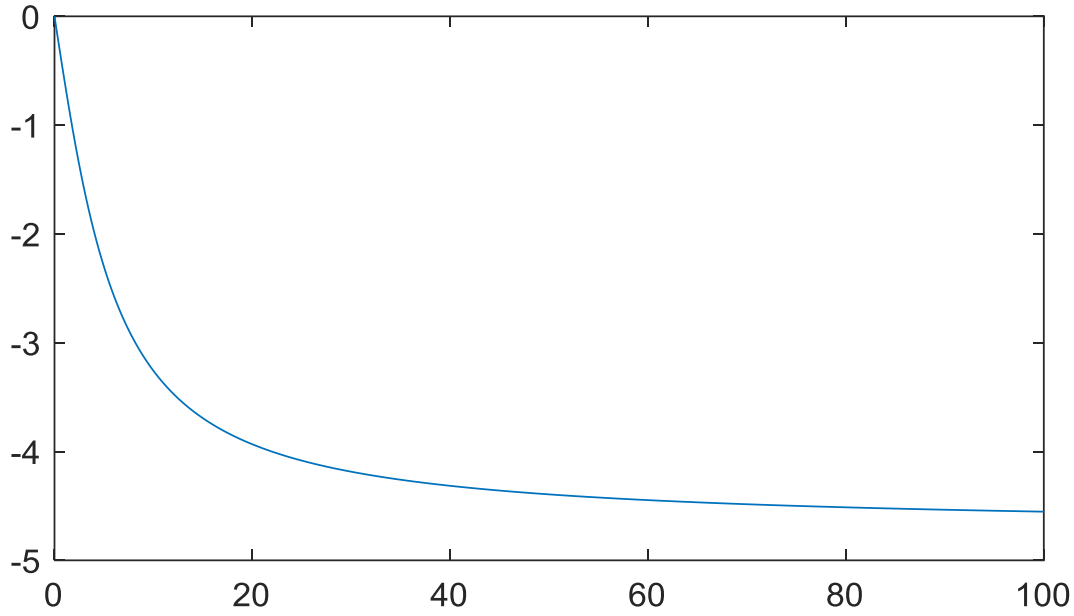
M function





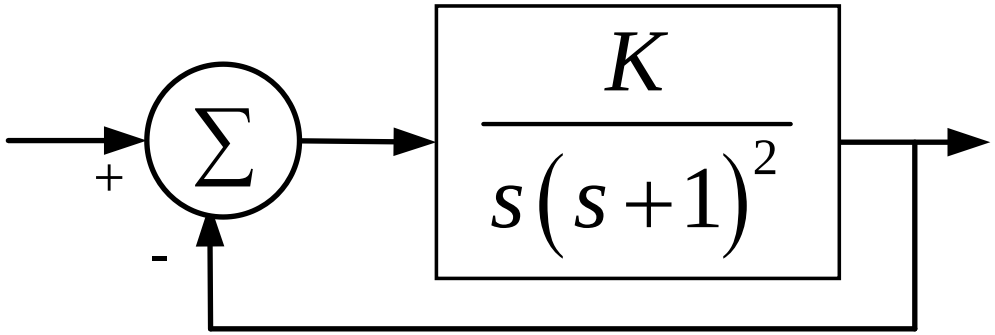
ϕ function





Application to Bode diagrams - 1

- If a system is



- Closed–Loop Transfer Function

$$\frac{\frac{K}{s(s+1)^2}}{1+\frac{K}{s(s+1)^2}} = \frac{K}{K+s(s+1)^2}$$

Numerical

$$\frac{K}{s(s+1)^2}$$

Frequency ω	**Phase (Degrees)**
0	-90
0.1	-101
1	-180
10	-258
∞	-270

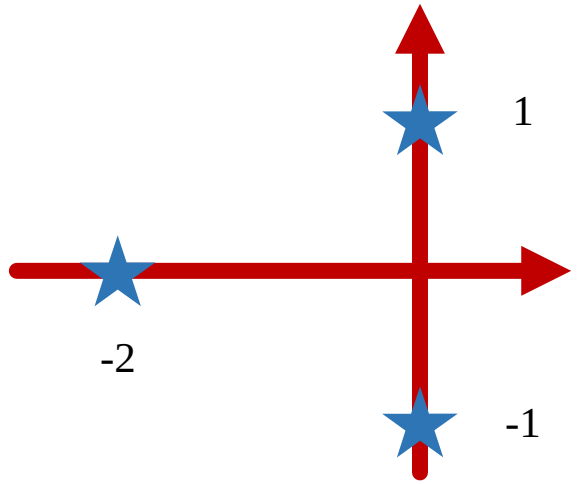
Application to Bode diagrams - 2

- Closed–loop characteristic equation $K + s(s+1)^2 = 0$

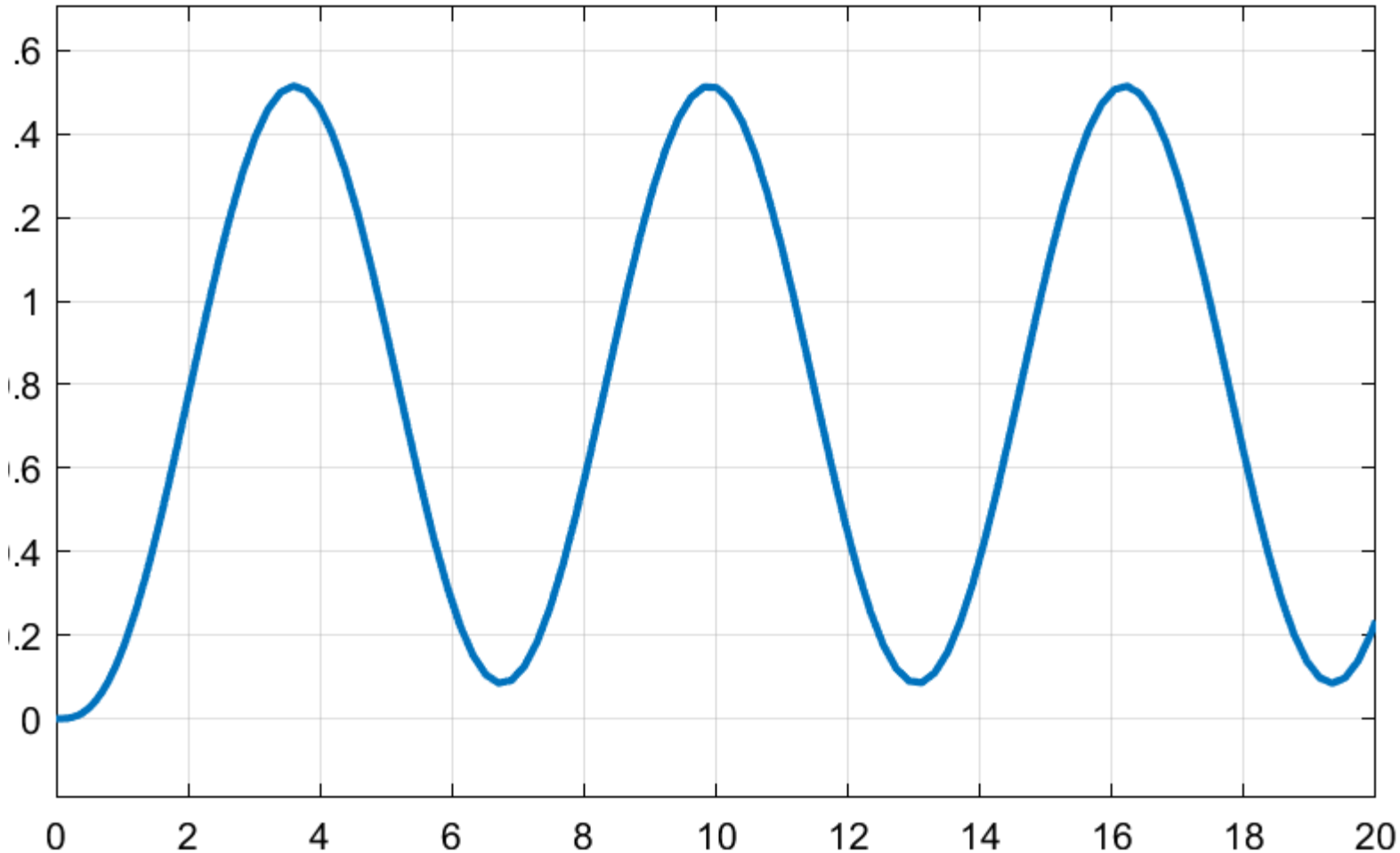
- When K=2, then $2 + s(s+1)^2 = 0 \longrightarrow s^3 + 2s^2 + s + 2 = 0$

  which can be factored to give: $(s+j)(s-j)(s+2) = 0$

- Two closed–loop poles on the imaginary axis, so for K=2 the system is marginally stable



$$H = \frac{2}{(s+2)(s^2+1)} = \frac{2}{5}\frac{1}{s+2} - \frac{2}{5}\frac{s}{s^2+1} + \frac{4}{5}\frac{1}{s^2+1}$$

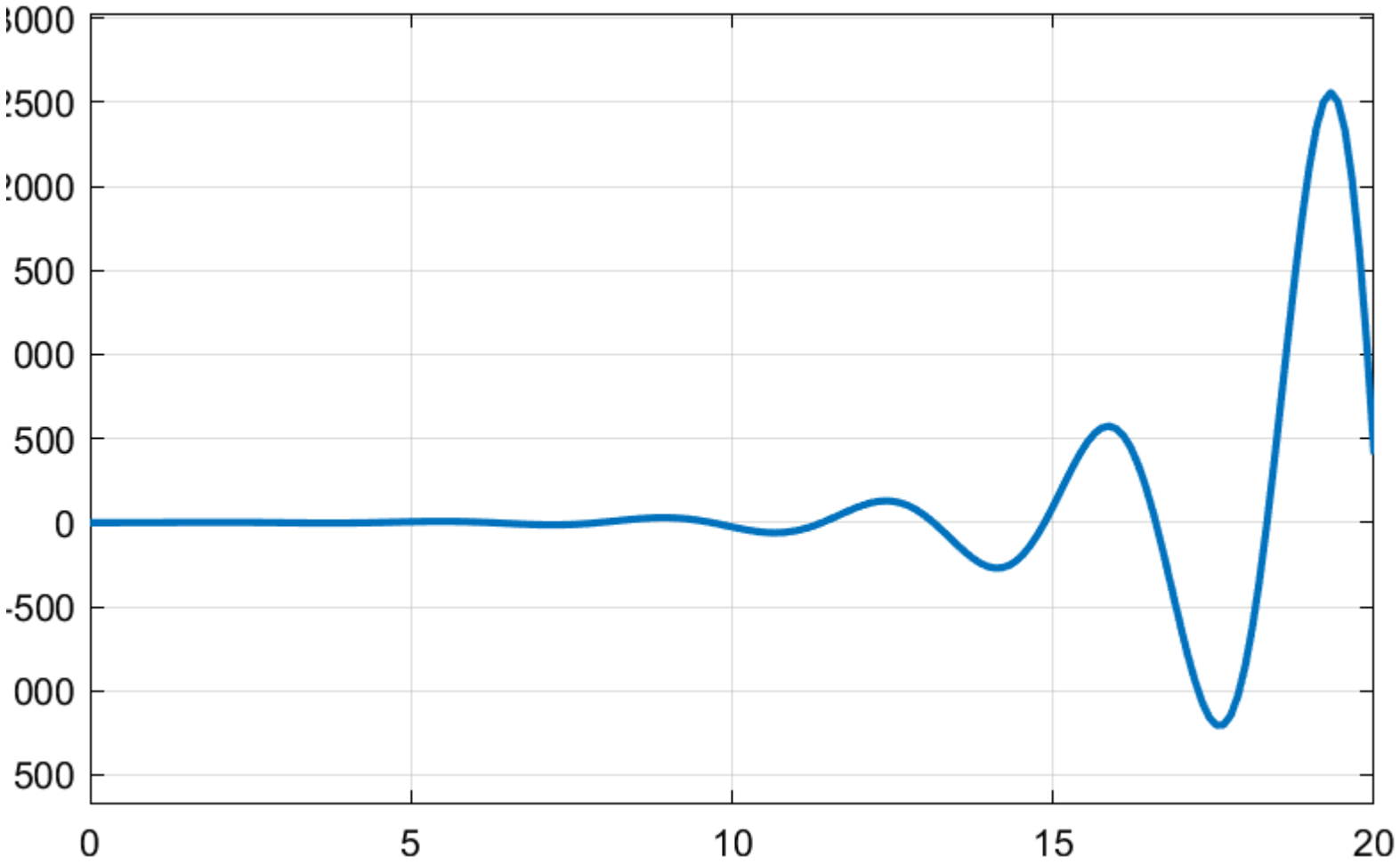

.6
.4
.2
1
.8
.6
.4
.2
0
0
2
4
6
8
10
12
14
16
18
20

Application to Bode diagrams - 3

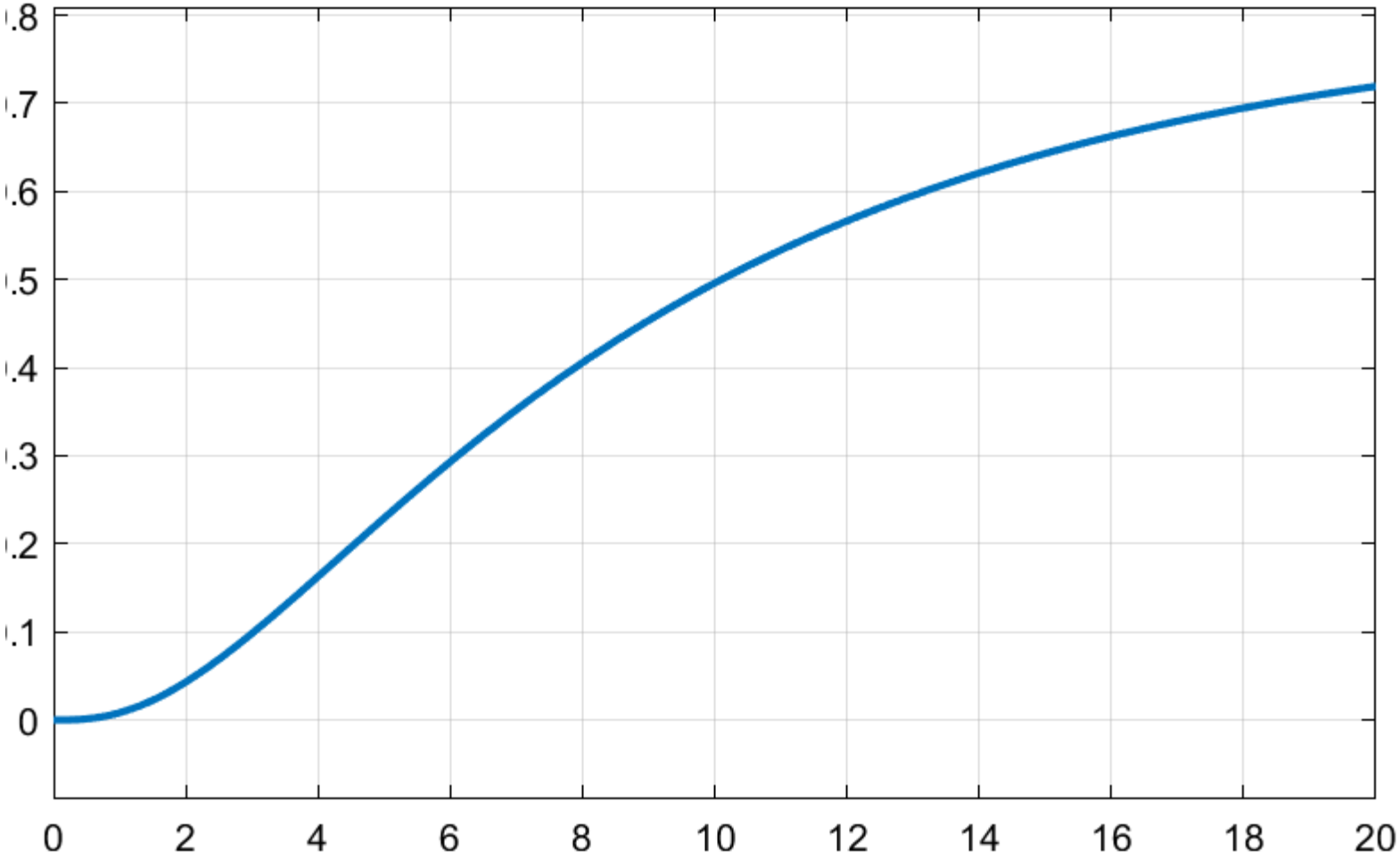
- When $K \neq 2$

$$s^3 + 2s^2 + s + K = 0$$

- When K=10

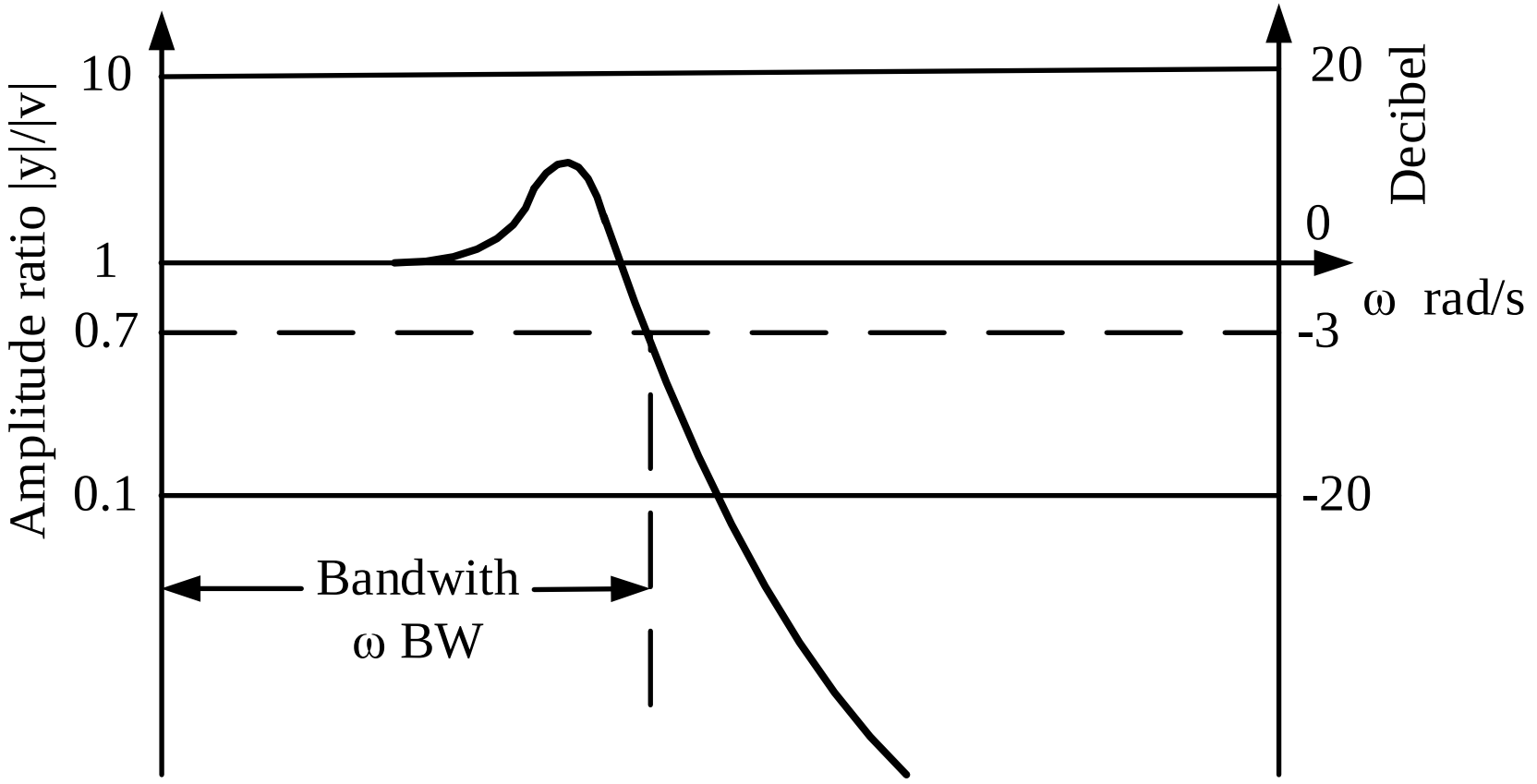


- When K=0.1



Bandwidth

- The bandwidth is the frequency on the log(M) plot at which the curve crosses the 3dB line. Measure of the speed of response, where a large bandwidth implies a faster response



Use of Frequency Response – 1

- Analogue control implementation
- Digital control implementation

Summary

- The steady state response of a system to a harmonic input is a harmonic with the same frequency but a different magnitude and phase – this is the frequency response
- The magnitude and phase shift of the output are both functions of frequency.
- The Bode Diagram consists of two parts, both plotted against the log–frequency, i.e. log–magnitude and phase angle
- Closed–loop stability is found from open–loop frequency response, and the control engineer can specify the required Gain Margin and Phase Margin